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Chiral thiophosphoramidate catalyzed asymmetric aryl transfer reactions for the synthesis of functional diarylmethanols

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ABSTRACT

In this investigation, chiral thiophosphoramidate **3d** was easily prepared from chiral (1*R*,2*R*)-1,2-diphenylethylenediamine and then applied as an efficient chiral ligand in the catalytic asymmetric arylation reactions of various aromatic aldehydes. The corresponding diarylmethanol products were produced with good to excellent yields (up to 98%) and enantioselectivities (up to 94%). The recovery of chiral ligand **3d** could be as high as 96%.

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1. Introduction

The catalytic enantioselective addition of aryl groups to aldehydes has attracted great attention.¹ The resulting chiral diaryl methanols are significant backbones of many biologically active natural products and medicine,² such as cizolirtine **I**,³ neobenodine **II**,^{2a,4} carbinoxamine **III**⁵ and orphenadrine **IV**.^{4,6} In addition, chiral diarylmethanols can be used to prepare many useful drugs **V**⁷ and **VI**.^{8,9} through nucleophilic displacement at the stereocenter without the loss of enantiomeric purity (Fig. 1).

The catalytic asymmetric addition of arylmetal reagents to aromatic aldehydes is been regarded as one of the most useful and straightforward methods for the synthesis of enantioenriched diarylmethanols by C–C bond formation.^{1,10} Over the past few decades, great progress has been achieved in the asymmetric addition of arylaluminiums,¹¹ aryl-Grignard reagents,¹² arylsilane reagents,¹³ aryl titanium reagents¹⁴ and aryl lithium reagents¹⁵ to aromatic aldehydes. Compared with the above developed methods, the asymmetric addition of arylzinc reagents^{1,16–18} has advantages of wide substituent tolerance and mild reaction conditions due to the relatively low reactivity of arylzinc reagents. In 2002, Bolm et al. developed a significant arylation method for the catalyzed asymmetric synthesis of chiral secondary diarylmethanols with excellent enantioselectivities.¹⁹ The active arylzinc species used in the asymmetric arylation, were generated in situ from a

boron-zinc transmetalation protocol. Since various aryl boronic acids with functional groups are commercially available or can be easily prepared, the scope of aryl boronic acids as aryl sources has been extensively explored and many efficient asymmetric protocols have been developed.^{20,21} Despite the pioneering studies that have been reported on the catalytic asymmetric addition of aryl zinc reagents to aromatic aldehydes, the development of highly efficient asymmetric catalyst systems with wide substrate tolerance using easily prepared chiral ligands is still desirable.

Chiral 1,2-diamino phosphoramidate ligands are very efficient in organozinc reagent catalytic asymmetric reactions,^{17m,20d,22–24} organocatalyzed asymmetric reactions²⁵ and enantiodiscrimination of chiral acids.²⁶ To explore the application of chiral phosphoramidate-Zn(II) ligands, we herein report examples of catalytic asymmetric arylation reactions from arylboronic acids and aromatic aldehydes for the synthesis of chiral secondary diarylmethanols.

2. Results and discussion

Systematic studies of a model reaction comprising 4-methyl benzaldehyde, phenyl boronic acid, dialkylzinc reagent and a chiral ligand, are summarized in Table 1. The phenylzinc reagent was prepared in situ by heating a mixture of dimethyl zinc and phenyl boronic acid in toluene for 12 h at 60 °C. The effect of the chiral ligand was first screened. As shown in Table 1, the corresponding diarylmethanol **4a** was formed with high yield and moderate enantioselectivity in the presence of quinine derived phosphoramidate chiral ligand **1** (Table 1, entry 1). The reaction using chiral (1*R*,2*R*)-1,2-diaminocyclohexane derived phosphoramidate **2** proceeded to give **4a** with 90% yield and 64% ee (Table 1, entry 2).

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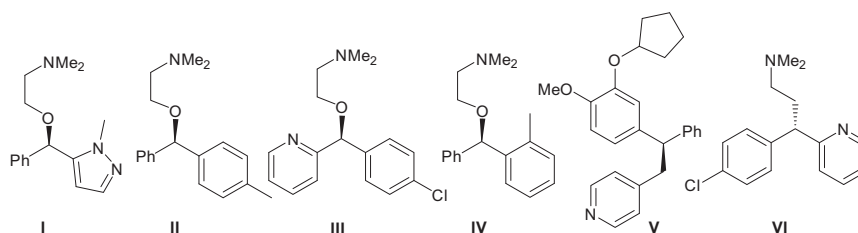
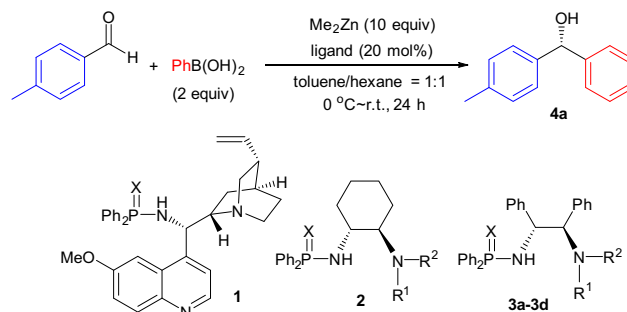


Figure 1. Structures of biologically active diarylmethanol derivatives.

Table 1

Catalytic asymmetric phenyl transfer from boronic acid to 4-methyl benzaldehyde using a phosphoramidate chiral ligand^a



Entry	Ligand	X	NR ¹ R ²	Yield ^b (%)	ee ^c (%)
1	1	O	—	89	55
2	2	O	NH ⁱ Pr	90	64
3	3a	O	NEt ₂	94	34
4	3b	O	NHBn	87	70
5	3c	O	NH ⁱ Pr	89	77
6	3d	S	NH ⁱ Pr	92	78

^a Unless otherwise noted, all reactions were carried out with benzaldehyde on a 0.5 mmol scale.

^b Isolated yields.

^c Determined by chiral GC analysis and the absolute configuration was assigned by comparison of the sign of specific rotation value with the literature value.

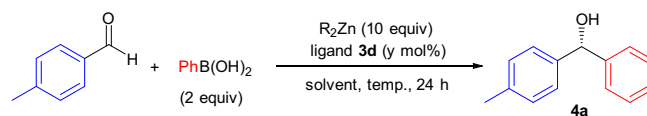
Chiral (1*R*,2*R*)-1,2-diphenylethylenediamine derived *N,N*-diethyl substituted phosphoramidate **3a** gave the diarylmethanol **4a** with excellent yield but only 34% *ee* (Table 1, entry 3). *N*-Benzyl monosubstituted phosphoramidate **3b** exhibited higher stereoselectivity for the asymmetric aryl transfer reaction (Table 1, entry 4). When *N*-isopropyl monosubstituted phosphoramidate **3c** was used, the diarylmethanol **4a** was afforded with 87% yield and 77% *ee* (Table 1, entry 5). Finally, *N*-isopropyl monosubstituted thiophosphoramidate **3d** proved to be the optimal chiral ligand, and under the same reaction conditions, afforded **4a** with 92% yield and 78% *ee* (Table 1, entry 6). As a result, chiral thiophosphoramidate **3d** was identified as the best chiral ligand for the catalytic asymmetric phenyl transfer reaction.

The reaction conditions, such as the solvent, reaction temperature, dialkyl zinc reagent and the amount of the chiral ligand were further optimized. The results are listed in Table 2. Various solvents, such as toluene, hexane, THF and MTBE, were also evaluated; the mixed solvent (toluene/hexane = 1:1) was found to be the best solvent giving 92% yield with 78% *ee* of the product (Table 2, entries 1–5). The effect of the temperature was then investigated. The yield and *ee* increased slightly when the reaction temperature was decreased (Table 2, entries 6 and 7). When decreasing the temperature to –40 °C, diarylmethanol **4a** was obtained with 85% yield and 82% *ee* (Table 2, entry 8). If the dimethylzinc reagent was replaced by diethylzinc reagent, the corresponding alcohol product was afforded with 90% yield and 83% *ee* (Table 2, entry 9). Increasing the loading of chiral ligand to 30 mol %, excellent enantioselectivity of the product was obtained

(93% *ee*, Table 2, entry 10). When 40 mol % of chiral ligand **3d** was used, almost no further improvement on yield and *ee* of the product was observed (Table 2, entry 11). Thus, the model reaction was best performed using 2 equiv of phenyl boronic acid and 10 equiv of Et₂Zn with 30 mol % chiral ligand **3d** in mixed solvent (toluene/hexane = 1:1) at –40 °C for 24 h. To achieve high enantioselectivity, high loading of the chiral ligand **3d** (30 mol %) was required, but it can be recovered by flash column chromatography without any loss of purity. The recovery of chiral ligand **3d** could be as high as 96%.

With the optimized conditions in hand, we next investigated the substrate scope and limitation of the asymmetric arylation reaction system using 30 mol % chiral ligands **3d**. The results are listed in Table 3.

The asymmetric arylation of a variety of aromatic aldehydes with in situ generated aryl zinc reagents took place, affording the anticipated chiral diarylmethanols products in good to excellent yields (81–98% yields) and with good to excellent enantioselectivity (82–94% *ee*) (Table 3, entries 1–10). Aromatic aldehydes with electron-withdrawing or electro-donating substituents at the *ortho*-, *meta*-, and *para*-positions of the benzene rings worked well. This catalytic asymmetric reaction system also worked well when 1-naphthaldehyde or 2-naphthaldehyde was used as the substrate (Table 3, entries 11 and 12). When 2-thiophenyaldehyde was used as substrate 90% yield and 51% *ee* were obtained (Table 3, entries 13). As shown in Table 3, entries 14–16, other aryl transferring reagents also reacted very smoothly, and the corresponding chiral diarylmethanols products could be obtained with good yields and

Table 2Optimization of the reaction conditions for the catalytic asymmetric phenyl transfer from boronic acid to 4-methyl benzaldehyde using chiral ligand **3d**^a

Entry	y (mol %)	R	Temp.	Solvent	Yield ^b (%)	ee ^c (%)
1 ^d	20	Me	0 °C–rt	Toluene/hexane	92	78
2	20	Me	0 °C–rt	Toluene	91	76
3	20	Me	0 °C–rt	Hexane	88	77
4 ^d	20	Me	0 °C–rt	Hexane/MTBE	86	75
5 ^d	20	Me	0 °C–rt	Hexane/THF	Trace	N.D. ^e
6 ^d	20	Me	–10 °C	Toluene/hexane	90	79
7 ^d	20	Me	–20 °C	Toluene/hexane	91	81
8 ^d	20	Me	–40 °C	Toluene/hexane	85	82
9 ^d	20	Et	–40 °C	Toluene/hexane	90	83
10 ^{d,f}	30	Et	–40 °C	Toluene/hexane	91	93
11 ^d	40	Et	–40 °C	Toluene/hexane	93	93

^a Unless otherwise noted, all reactions were carried out with benzaldehyde on a 0.5 mmol scale.^b Isolated yields.^c Determined by chiral GC analysis and the absolute configuration was assigned by comparison of the sign of specific rotations with the literature values.^d The ratio of the solvent volume was 1:1.^e Not determined.^f The chiral ligand can be recovered with 96% yield.

excellent enantioselectivities. Several functionalized aryl boronic acids were used as aryl transfer reagents in the catalytic asymmetric addition to various aromatic aldehydes with functional groups. Excellent yields (90–95% yields) and good to excellent enantioselectivities (80–92% ee) could be attained (Table 3, entries 17–22). It is noteworthy that the ester group can be tolerated in this catalytic asymmetric aryl transfer system (entry 21).

Compared with aromatic aldehydes, aliphatic aldehydes are less active and enantioselective. Thus, many excellent chiral ligands, which are highly efficient to aromatic aldehydes, are less effective towards aliphatic aldehydes.²⁷ When 3-methylbutyraldehyde and phenyl boronic acid were used as the substrate and aryl transfer reagent respectively, the corresponding product (*S*)-3-methyl-1-phenylbutan-1-ol was obtained with 79% yield and 55% ee under the optimized reaction conditions.

3. Conclusions

In conclusion, thiophosphoramidate **3d** was easily prepared from chiral 1, 2-diphenylethylenediamine and was identified as an efficient chiral ligand in the catalytic asymmetric arylation of aldehydes. Using this protocol, various chiral functionalized diaryl-methanols with high enantioselectivities could be prepared conveniently from commercially available aryl boronic acids and aldehydes.

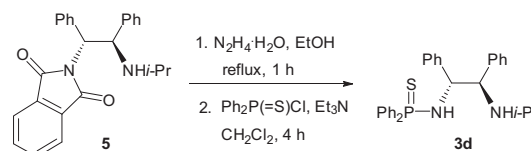
4. Experimental

4.1. General

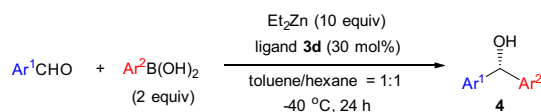
All experiments were carried out in dried glassware with magnetic stirring under an atmosphere of dry nitrogen. ¹H NMR (400 MHz) and ¹³C NMR (100 MHz) spectra were recorded in CDCl₃ solutions using a 400 MHz spectrometer. Chemical shifts are reported in parts per million (ppm, δ) relative to CDCl₃ (δ 7.26 for ¹H NMR), or CDCl₃ (δ 77.0 for ¹³C NMR). Multiplicities are indicated as s (singlet), d (doublet), t (triplet), q (quartet), m (multiplet) and br (broad). Commercial reagents were used as received unless otherwise indicated. All solvents were purified and dried prior to use according to standard methods. Optical rotations were mea-

sured on a polarimeter and specific rotations are reported as follows: [α]_D²⁵ (c g/100 mL, solvent). GC analysis was performed on a gas chromatograph with a FID detector on fused silica chiral capillary column (Chirasil Dex CB column, 25 m Length × 0.32 mm ID × 0.25 mm film thickness). The high performance liquid chromatography (HPLC) analysis was performed using Chiralcel OD-H, AD-H, OJ-H or AS-H column. Chiral ligands **1**,^{23f} **2**,^{23h} **3a**,^{23c} **3b**,^{23b} and **3c**^{23b} were prepared according to the literature values. Compound **5**^{23c} was synthesized according to reported method.

4.2. Preparation of chiral ligand **3d**

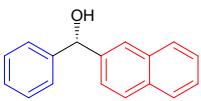
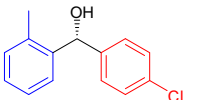
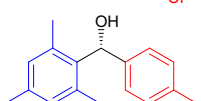
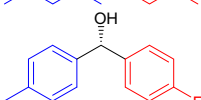
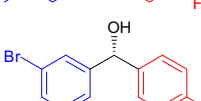
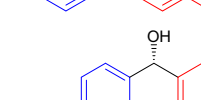
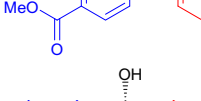


To a solution of **5** (845 mg, 2.2 mmol) in ethanol (3 mL), hydrazine monohydrate (0.97 mL, 20 mmol) was added, and the reaction mixture was stirred at reflux (oil bath, 85 °C) for 2 h. After the resultant mixture was cooled to room temperature, diethyl ether (30 mL) and CH₂Cl₂ (30 mL) were added to form precipitates, which were removed by filtration. The filtrate was dried over anhydrous MgSO₄, and the solvent was removed under reduced pressure to afford 508 mg of white solid, which was used in the next step without further purification. A reaction mixture of the white solid in CH₂Cl₂ (15 mL) with Et₃N (556 mg, 5.5 mmol) was stirred at room temperature for 15 min. After being cooled to 0 °C, it was added to a solution of diphenylthiophosphinic chloride (1.26 g, 5.0 mmol) in dry CH₂Cl₂ (15 mL) dropwise and stirred for 30 min at 0 °C. The temperature was allowed to warm to room temperature and then kept at room temperature for approximately 5 h. After the reaction mixture was cooled to 0 °C, water (30 mL) and CH₂Cl₂ (30 mL) were added. The organic layer was dried over anhydrous MgSO₄ and concentrated. The crude product was sub-

Table 3
Asymmetric arylation of aromatic aldehydes^a

Entry	Product	Name	Yield ^b (%)	ee ^c (%)
1		(R)- 4a	91	93
2		(R)- 4b	95	84
3		(R)- 4c	98	84
4		(R)- 4d	93	94
5		(R)- 4e	92	92
6		(R)- 4f	96	84
7		(R)- 4g	88	90
8		(R)- 4h	81	93
9		(R)- 4i	83 (82 ^d)	87 (81 ^d)
10		(R)- 4j	95 (93 ^d)	90 (84 ^d)
11		(R)- 4k	86	82
12		(R)- 4l	82	87
13		(R)- 4m	90	51
14		(S)- 4a	85	91
15		(S)- 4g	92	89

Table 3 (continued)

Entry	Product	Name	Yield ^b (%)	ee ^c (%)
16		(S)- 4l	84	86
17		(R)- 4n	90	91
18		(R)- 4o	92	92
19		(S)- 4p	91	90
20		(R)- 4q	90	89
21		(S)- 4r	93	80
22		(R)- 4s	95	92

^a Unless otherwise noted, all reactions were carried out with benzaldehyde on a 0.5 mmol scale.

^b Isolated yields.

^c Determined by chiral GC or HPLC analysis.

^d 1.3 equiv PhB(OH)₂, 6.5 equiv Et₂Zn and 15 mol % chiral ligand **3d** were used.

jected to silica gel column chromatography (EtOAc/hexane = 3/1), which afforded 703 mg (68%) of **3d** as a white solid. Mp, 162–163 °C; $[\alpha]_D^{30.9} = +4.9$ (c 1.00, CH₂Cl₂); ¹H NMR (400 MHz, CDCl₃): δ 0.97 (t, *J* = 6.4 Hz, 6H), 1.64 (s, 1H), 2.55–2.68 (m, 1H), 3.93 (d, *J* = 8.2 Hz, 1H), 4.47–4.57 (m, 1H), 4.62 (d, *J* = 4.0 Hz, 1H), 6.87–7.27 (m, 13H), 7.36–7.54 (m, 3H), 7.61–7.76 (m, 2H), 7.81–7.98 (m, 2H); ¹³C NMR (100 MHz, CDCl₃): δ 21.9, 24.4, 45.3, 61.2, 65.7 (d, *J* = 8.0 Hz), 126.8, 127.2, 127.50, 127.54, 127.7, 128.0, 128.1, 128.27, 128.34, 128.5, 131.0, 131.1, 131.5, 132.1, 132.2, 132.7, 133.8, 134.7, 135.8, 140.2, 140.7; ³¹P NMR (162 MHz, CDCl₃): δ 60.5; HRMS (ESI): (*m/z*) [M+H]⁺ calcd for C₂₉H₃₂N₂SP, 471.1946; found, 471.1945.

4.3. General procedure for the catalytic asymmetric arylation of aldehyde

Under an N₂ atmosphere, to a reaction tube which contained arylboronic acid (1.0 mmol) and toluene (2 mL) was added diethylzinc (5.0 mmol, 1.0 M in hexane) at room temperature, and then heated at 60 °C for 12 h. After the flask was cooled to rt, the reaction mixture was added to another schlenk tube, which contained the chiral ligand (0.15 mmol) and toluene (3 mL), and the resulting mixture was stirred for 30 min. Aldehyde (0.5 mmol) was then added immediately to the mixture at –40 °C. The mixture was stirred for 24 h at –40 °C and quenched by adding aqueous HCl (1.0 M, 5 mL). Extraction with EtOAc (10 mL × 3) gave the combined organic layers, which were washed with brine (10 mL), dried over MgSO₄ and concentrated in vacuo to give a residue, which was subjected to silica gel column chromatography (EtOAc/hexane = 1/10), affording the corresponding enantioen-

riched secondary alcohol. The ee value of the secondary alcohol was determined by chiral GC or HPLC. The absolute configuration of the alcohol was assigned as (R) or (S) by comparison of the specific rotation data with the reported literature data.

4.3.1. (4-Methylphenyl)(phenyl)methanol **4a**^{15d}

91% yield and 93% ee were obtained. $[\alpha]_D^{30} = +3.8$ (c 1.0, CHCl₃) (Lit.^{15d} $[\alpha]_D^{25} = -5$ (c 1.0, CHCl₃) for 94% ee (S)); the ee value was determined by Chiral GC Chirasil Dex CB [Column temperature: 135 °C, *t_R* = 95.3 min (R), *t_R* = 102.0 min (S)]. ¹H NMR (400 MHz, CDCl₃): δ 2.42 (s, 3H), 3.15 (br, 1H), 5.80 (s, 1H), 7.18–7.24 (m, 2H), 7.29–7.36 (m, 3H), 7.37–7.46 (m, 4H); ¹³C NMR (100 MHz, CDCl₃): δ 21.2, 76.1, 126.5, 127.5, 128.5, 129.2, 137.3, 141.0, 144.0.

4.3.2. 2-Methoxyphenyl(phenyl)methanol **4b**²⁸

95% yield and 84% ee were obtained. $[\alpha]_D^{25.8} = +16.7$ (c 1.0, CHCl₃) (Lit.²⁸ $[\alpha]_D^{20} = +22.8$ (c 1.27, CHCl₃) for 70% ee (R)); the ee value was determined on a Daicel Chiralcel OD-H column [(10% IPA/hexane, 1.0 mL/min, retention time: 9.2 min (minor) and 10.2 min (major)], *t_R* = 9.2 min (S), *t_R* = 10.2 min (R)]. ¹H NMR (400 MHz, CDCl₃): δ 3.22 (d, *J* = 4.2 Hz, 1H), 3.84 (s, 3H), 6.12 (s, 1H), 6.90–7.05 (m, 2H), 7.27–7.49 (m, 7H); ¹³C NMR (100 MHz, CDCl₃): δ 55.5, 72.2, 110.7, 121.1, 123.8, 126.6, 127.2, 127.8, 128.2, 128.7, 132.2, 136.1, 143.5, 149.8, 156.7.

4.3.3. 4-Methoxyphenyl(phenyl)methanol **4c**^{15d}

98% yield and 84% ee were obtained, $[\alpha]_D^{26.3} = +6.6$ (c 1.0, CHCl₃) (Lit.^{15d} $[\alpha]_D^{25} = -19$ (c 1.0, CHCl₃) for 92% ee (S)); the ee value was determined on a Daicel Chiralcel AD-H column [(10% IPA/hexane, 0.8 mL/min, retention time: 14.0 min (major) and 15.0 min

(minor), $t_R = 14.0$ min (R), $t_R = 15.0$ min (S)]. ^1H NMR (400 MHz, CDCl_3): δ 2.32 (d, $J = 3.1$ Hz, 1H), 3.81 (s, 3H), 5.83 (s, 1H), 6.85–6.96 (m, 2H), 7.23–7.495 (m, 7H); ^{13}C NMR (100 MHz, CDCl_3): δ 55.3, 75.8, 113.9, 126.4, 127.2, 127.5, 127.9, 128.5, 136.2, 144.0, 159.0.

4.3.4. (2-Methylphenyl)(phenyl)methanol 4d²⁹

93% yield and 94% ee were obtained. $[\alpha]_D^{25.1} = +6.3$ (c 1.0, CHCl_3) (Lit.²⁹ $[\alpha]_D^{20} = +6.8$ (c 1.0, CHCl_3) for 97% ee (R)). The ee value was determined on a Daicel Chiralcel OD-H column [(10% IPA/hexane, 1.0 mL/min, retention time: 7.5 min (major) and 7.9 min (minor)), $t_R = 7.5$ min (R), $t_R = 7.9$ min (S)]. ^1H NMR (400 MHz, CDCl_3): δ 2.19 (br, 1H), 2.28 (s, 3H), 6.03 (s, 1H), 7.12–7.61 (m, 9H); ^{13}C NMR (100 MHz, CDCl_3): δ 19.4, 73.4, 126.1, 126.3, 127.1, 127.5, 127.6, 128.5, 130.5, 135.4, 141.5, 142.9.

4.3.5. (3-Bromophenyl)(phenyl)methanol 4e²⁰ⁱ

92% yield and 92% ee were obtained. $[\alpha]_D^{28.1} = -11.0$ (c 2.0, CHCl_3) (Lit.²⁰ⁱ $[\alpha]_D^{25} = +25.7$ (c 0.944, CHCl_3) for 87% ee (S)); the ee value was determined on a Daicel Chiralcel AS-H column [(8% IPA/hexane, 1 mL/min, retention time: 7.0 min (minor) and 7.3 min (major)), $t_R = 7.0$ min (S), $t_R = 7.3$ min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 2.56 (s, 1H), 5.79 (s, 1H), 7.18–7.45 (m, 8H), 7.53–7.63 (m, 1H); ^{13}C NMR (100 MHz, CDCl_3): δ 75.6, 115.3, 120.7, 122.6, 125.1, 126.6, 128.0, 128.8, 129.6, 130.1, 130.6, 143.2, 146.0.

4.3.6. (4-Bromophenyl)(phenyl)methanol 4f^{15d}

96% yield and 84% ee were obtained. $[\alpha]_D^{26.5} = -15.2$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{25} = +19$ (c 1.0, CHCl_3) for 91% ee (S)); the ee value was determined on a Daicel Chiralcel AD-H column [(5% IPA/hexane, 0.8 mL/min, retention time: 16.1 min (major) and 17.7 min (minor)), $t_R = 16.1$ min (R), $t_R = 17.7$ min (S)]. ^1H NMR (400 MHz, CDCl_3): δ 3.64 (s, 1H), 5.65 (s, 1H), 7.17–7.25 (m, 2H), 7.31–7.43 (m, 5H), 7.45–7.52 (m, 2H); ^{13}C NMR (100 MHz, CDCl_3): δ 75.5, 121.4, 126.7, 127.9, 128.4, 128.7, 131.6, 142.8, 143.4.

4.3.7. (4-Chlorophenyl)(phenyl)methanol 4g^{15d}

88% yield and 90% ee were obtained. $[\alpha]_D^{27.4} = -5.3$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{25} = +19$ (c 1.0, CHCl_3) for 92% ee (S)); the ee value was determined on a Daicel Chiralcel AD-H column [(5% IPA/hexane, 0.8 mL/min, retention time: 17.9 min (minor) and 18.5 min (major)), $t_R = 17.9$ min (S), $t_R = 18.5$ min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 3.68 (s, 1H), 5.67 (s, 1H), 7.26–7.33 (m, 2H), 7.34–7.49 (m, 7H); ^{13}C NMR (100 MHz, CDCl_3): δ 75.6, 126.6, 127.9, 128.6, 128.7, 133.3, 142.4, 143.4.

4.3.8. (3-Fluorophenyl)(phenyl)methanol 4h³⁰

81% yield and 93% ee were obtained. $[\alpha]_D^{26.6} = -14.3$ (c 1.0, CHCl_3); the ee value was determined on a Daicel Chiralcel OD-H column [(5% IPA/hexane, 0.8 mL/min, retention time: 17.8 min (minor) and 18.4 min (major)), $t_R = 17.8$ min (S), $t_R = 18.4$ min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 3.54 (s, 1H), 5.74 (s, 1H), 6.97–7.10 (m, 1H), 7.12–7.26 (m, 2H), 7.28–7.54 (m, 6H); ^{13}C NMR (100 MHz, CDCl_3): δ 75.7, 113.3, 113.5, 114.2, 114.5, 122.0, 126.6, 128.0, 128.7, 129.9, 130.0, 143.3, 146.3, 161.7, 164.2. ^{19}F CPD NMR (376 MHz, CDCl_3): δ -112.71.

4.3.9. (4-Fluorophenyl)(phenyl)methanol 4i^{15d}

83% yield and 87% ee were obtained. $[\alpha]_D^{27.5} = -6.3$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{25} = +8$ (c 1.0, CHCl_3) for 92% ee (S)); the ee value was determined on a Daicel Chiralcel AS-H column [(8% IPA/hexane, 0.8 mL/min, retention time: 8.7 min (minor) and 9.0 min (major)), $t_R = 8.7$ min (S), $t_R = 9.0$ min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 2.33 (d, $J = 3.6$ Hz, 1H), 5.85 (d, $J = 3.0$ Hz, 1H), 6.90–

7.70 (m, 9H); ^{13}C NMR (100 MHz, CDCl_3): δ 75.6, 115.2, 115.4, 126.5, 127.8, 128.2, 128.3, 128.6, 139.6, 143.7, 160.9, 163.4. ^{19}F CPD NMR (376 MHz, CDCl_3): δ -115.04.

4.3.10. (4-(Trifluoromethyl)phenyl)phenylmethanol 4j^{15d}

95% yield and 90% ee were obtained. $[\alpha]_D^{30.7} = -23.1$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{25} = +27$ (c 1.0, CHCl_3) for 90% ee (S)); the ee value was determined by Chiral HPLC analysis with a Chiralcel OJ-H column [(hexane/2-propanol = 90/10, 0.8 mL/min, 220 nm); retention time: 12.7 min (major) and 13.2 min (minor)], $t_R = 12.7$ min (R), $t_R = 13.2$ min (S)]. ^1H NMR (400 MHz, CDCl_3): δ 3.22 (s, 1H), 5.79 (m, 1H), 7.31–7.43 (m, 5H), 7.45–7.56 (m, 2H), 7.57–7.73 (m, 2H); ^{13}C NMR (100 MHz, CDCl_3): δ 75.7, 124.2 (q, $J = 272.1$ Hz), 125.4 (q, $J = 3.8$ Hz), 126.69, 126.71, 128.1, 128.8, 129.6 (q, $J = 32.4$ Hz), 143.1, 147.5. ^{19}F NMR (376 MHz, CDCl_3): δ -62.4 (s).

4.3.11. Naphthalen-1-yl(phenyl)methanol 4k^{15d}

86% yield and 82% ee were obtained. $[\alpha]_D^{30.5} = +25.6$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{25} = -42$ (c 1.0, CHCl_3) for 96% ee (S)); the ee value was determined by Chiral HPLC analysis with a Chiralcel OJ-H column [(hexane/2-propanol = 80/20, 0.8 mL/min, 220 nm); retention time: 13.4 min (minor) and 16.9 min (major)], $t_R = 13.4$ min (S), $t_R = 16.9$ min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 3.00 (s, 1H), 6.48 (s, 1H), 7.27–7.61 (m, 8H), 7.63–7.72 (m, 1H), 7.83–8.00 (m, 2H), 8.02–8.13 (m, 1H); ^{13}C NMR (100 MHz, CDCl_3): δ 73.6, 124.1, 124.7, 125.4, 125.7, 126.2, 127.2, 127.7, 128.5, 128.6, 128.9, 129.6, 130.8, 134.0, 138.9, 143.2.

4.3.12. Naphthalen-2-yl(phenyl)methanol 4l^{15d}

82% yield and 87% ee were obtained. $[\alpha]_D^{30.8} = -5.5$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{25} = +6$ (c 1.0, CHCl_3) for 90% ee (S)); the ee value was determined by Chiral HPLC analysis with a Chiralcel OD-H column [(hexane/2-propanol = 90/10, 0.8 mL/min, 220 nm); retention time: 16.3 min (minor) and 19.3 min (major)], $t_R = 16.3$ min (S), $t_R = 19.3$ min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 3.00 (s, 1H), 6.00 (s, 1H), 7.28–7.64 (m, 8H), 7.77–8.00 (m, 4H); ^{13}C NMR (100 MHz, CDCl_3): δ 76.3, 124.9, 125.2, 126.1, 126.2, 126.8, 127.7, 127.8, 128.2, 128.4, 128.6, 133.0, 133.3, 143.7.

4.3.13. Phenyl(thiophen-2-yl)methanol 4m^{15d}

90% yield and 51% ee were obtained. $[\alpha]_D^{21.0} = -11.5$ (c 1.0, CHCl_3) (Lit.^{15d} $[\alpha]_D^{24} = +27$ (c 1.0, CHCl_3) for 96% ee (S)); the ee value was determined by Chiral HPLC analysis with a Chiralcel OD-H column [(hexane/2-propanol = 92/8, 0.8 mL/min, 220 nm); retention time: 10.4 min (minor) and 10.8 min (major)], $t_R = 10.4$ min (R), $t_R = 10.8$ min (S)]. ^1H NMR (400 MHz, CDCl_3): δ 3.78 (d, $J = 2.6$ Hz, 1H), 6.00 (s, 1H), 6.89–7.10 (m, 2H), 7.29–7.59 (m, 6H); ^{13}C NMR (100 MHz, CDCl_3): δ 72.3, 125.1, 125.5, 126.6, 126.8, 128.0, 128.6, 143.4, 148.4.

4.3.14. (4-Chlorophenyl)(o-tolyl)methanol 4n³¹

90% yield and 91% ee were obtained. $[\alpha]_D^{23.8} = -5.6$ (c 1.0, CHCl_3); the ee value was determined on a Daicel Chiralcel AD-H column [(5% IPA/hexane, 1.0 mL/min, retention time: 10.30 min (major) and 10.79 min (minor)), $t_R = 10.30$ min (R), $t_R = 10.79$ min (S)]. ^1H NMR (400 MHz, CDCl_3): δ 2.27 (s, 3H), 2.29 (s, 1H), 5.99 (d, $J = 3.4$ Hz, 1H), 7.16–7.21 (m, 1H), 7.22–7.35 (m, 6H), 7.44–7.52 (m, 1H); ^{13}C NMR (100 MHz, CDCl_3): δ 19.4, 72.8, 126.29, 126.33, 127.8, 128.5, 128.6, 130.7, 133.3, 135.4, 141.3.

4.3.15. Mesityl(p-tolyl)methanol 4o³²

92% yield and 92% ee were obtained. $[\alpha]_D^{24.0} = +39.0$ (c 1.0, CHCl_3); the ee value was determined on a Daicel Chiralcel OD-H column [(10% IPA/hexane, 1.0 mL/min, retention time: 5.32 min (minor) and 5.82 min (major)), $t_R = 5.32$ min (S), $t_R = 5.82$ min

(R)]. ^1H NMR (400 MHz, CDCl_3): δ 2.28 (s, 3H), 2.30 (s, 6H), 2.34 (s, 3H), 2.39 (s, 3H), 6.34 (d, J = 3.8 Hz, 1H), 6.92 (s, 2H), 7.09–7.28 (m, 4H); ^{13}C NMR (100 MHz, CDCl_3): δ 20.6, 20.9, 21.1, 71.0, 125.5, 128.9, 130.1, 136.1, 136.6, 137.1, 137.2, 140.1.

4.3.16. (4-Fluorophenyl)(*p*-tolyl)methanol 4p³³

91% yield and 90% *ee* were obtained, $[\alpha]_{\text{D}}^{22.8}$ = +6.2 (c 1.0, CHCl_3); the *ee* value was determined on a Daicel Chiralcel OD-H column [(5% IPA/hexane, 0.8 mL/min, retention time: 10.03 min (major) and 11.50 min (minor)), t_{R} = 10.03 min (S), t_{R} = 11.50 min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 2.40 (s, 3H), 2.88 (s, 1H), 5.74 (s, 1H), 6.99–7.10 (m, 2H), 7.16–7.22 (m, 2H), 7.23–7.28 (m, 2H), 7.30–7.39 (m, 2H); ^{13}C NMR (100 MHz, CDCl_3): δ 21.2, 75.4, 115.2 (d, J = 21.4 Hz), 126.5, 128.2 (d, J = 8.1 Hz), 129.3, 137.5, 139.8 (d, J = 3.0 Hz), 140.8, 162.1 (d, J = 245.0 Hz); ^{19}F NMR (376 MHz, CDCl_3): δ –115.2 (m).

4.3.17. (3-Bromophenyl)(4-fluorophenyl)methanol 4q^{12a}

89% yield and 89% *ee* were obtained, $[\alpha]_{\text{D}}^{22.9}$ = –8.2 (c 1.0, CHCl_3); the *ee* value was determined on a Daicel Chiralcel OD-H column [(5% IPA/hexane, 0.8 mL/min, retention time: 14.47 min (minor) and 17.28 min (major)), t_{R} = 14.47 min (S), t_{R} = 17.28 min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 3.46 (s, 1H), 5.67 (s, 1H), 6.96–7.07 (m, 2H), 7.14–7.35 (m, 4H), 7.38–7.47 (m, 1H), 7.52 (s, 1H); ^{13}C NMR (100 MHz, CDCl_3): δ 74.9, 115.5 (d, J = 21.4 Hz), 122.7, 125.1, 128.3 (d, J = 8.1 Hz), 129.4, 130.2, 130.8, 138.2 (d, J = 3.0 Hz), 145.8, 162.3 (d, J = 246.5 Hz); ^{19}F NMR (376 MHz, CDCl_3): δ –114.1 (m).

4.3.18. Methyl 4-((4-fluorophenyl)(hydroxy)methyl)benzoate 4r^{20m}

90% yield and 80% *ee* were obtained, $[\alpha]_{\text{D}}^{22.5}$ = –11.4 (c 1.0, CHCl_3); the *ee* value was determined on a Daicel Chiralcel OD-H column [(4% IPA/hexane, 1.0 mL/min, retention time: 23.78 min (major) and 25.54 min (minor)), t_{R} = 23.78 min (S), t_{R} = 25.54 min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 3.06 (s, 9H), 3.89 (s, 1H), 5.83 (s, 1H), 6.95–7.05 (m, 2H), 7.26–7.35 (m, 2H), 7.39–7.46 (m, 2H), 7.97–8.01 (m, 2H); ^{13}C NMR (100 MHz, CDCl_3): δ 52.2, 75.1, 115.4 (d, J = 21.4 Hz), 126.3, 128.4 (d, J = 8.2 Hz), 129.2, 129.8, 139.1 (d, J = 3.0 Hz), 148.8, 162.2 (d, J = 246.4 Hz), 167.1; ^{19}F NMR (376 MHz, CDCl_3): δ –114.4 (m).

4.3.19. (4-*tert*-Butylphenyl)(naphthalen-3-yl)methanol 4s³⁴

95% yield and 92% *ee* were obtained, $[\alpha]_{\text{D}}^{22.2}$ = –8.8 (c 1.0, CHCl_3); the *ee* value was determined on a Daicel Chiralcel AD-H column [(6% IPA/hexane, 1.0 mL/min, retention time: 12.25 min (minor) and 13.45 min (major)), t_{R} = 12.25 min (S), t_{R} = 13.45 min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 1.32 (s, 9H), 2.37 (s, 1H), 6.02 (s, 1H), 7.33–7.43 (m, 4H), 7.45–7.55 (m, 3H), 7.78–7.99 (m, 4H); ^{13}C NMR (100 MHz, CDCl_3): δ 31.3, 34.5, 76.2, 124.81, 124.85, 125.5, 125.9, 126.2, 126.5, 127.7, 128.1, 128.3, 132.9, 133.3, 140.8, 141.2.

4.3.20. 3-Methyl-1-phenylbutan-1-ol³⁵

79% yield and 55% *ee* were obtained, $[\alpha]_{\text{D}}^{22.3}$ = –11.3 (c 1.0, $\text{CH}_3\text{CH}_2\text{OH}$); the *ee* value was determined by Chiral GC Chirasil Dex CB [Column temperature: 140 °C, t_{R} = 7.65 min (S), t_{R} = 8.09 min (R)]. ^1H NMR (400 MHz, CDCl_3): δ 0.97 (d, J = 2.4 Hz, 3H), 0.99 (d, J = 2.3 Hz, 3H), 1.47–1.59 (m, 1H), 1.68–1.83 (m, 2H), 1.99 (s, 1H), 4.72–4.79 (m, 1H), 7.26–7.34 (m, 1H), 7.34–7.40 (m, 4H); ^{13}C NMR (100 MHz, CDCl_3): δ 22.3, 23.1, 24.8, 48.4, 72.8, 125.9, 127.5, 128.5, 145.2.

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A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tetasy.2016.11.011>.

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